Name: Ayiik Mayen

Date: October 11, 2026

Course/Class Title: Practical Development Setup Assignment

Assignment: Practical Development Setup

Section 1: Task 1 - Flutter & Dart Setup

---

1. Flutter Doctor Output

```
C:\Users\Admin>flutter doctor

Doctor summary (to see all details, run flutter doctor -v):
[!] Flutter (Channel stable, 3.47.1, on Microsoft Windows [Version
10.0.26200.9168], locale en-KE)
    ! Warning: `dart` on your path resolves to C:\dart-sdk\bin\dart.exe, which is
not inside your current Flutter SDK checkout at C:\flutter. Consider adding
C:\flutter\bin to the front of your path.
[√] Windows Version (Windows 11 or higher, 25H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
    X cmdline-tools component is missing.
    Try installing or updating Android Studio.
    Alternatively, download the tools from
https://developer.android.com/studio#command-line-tools-only and make sure to set
the ANDROID_HOME environment variable.
    See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
    Run `flutter doctor --android-licenses` to accept the SDK licenses.
    See https://flutter.dev/to/windows-android-setup for more details.
[√] Chrome - develop for the web
[√] Visual Studio - develop Windows apps (Visual Studio Build Tools 2026 18.9.0)
[√] Connected device (3 available)
[√] Network resources

! Doctor found issues in 2 categories.
```

2. App Initialization & Target Devices

```
Admin@DESKTOP-LOGVC1K MINGW64 ~/documents
$ flutter create my_first_app
Creating project my_first_app...
Resolving dependencies in my_first_app...
Got dependencies in my_first_app.
Wrote 129 files.

All done!
In order to run your application, type:

  $ cd my_first_app
  $ flutter run

Your application code is in my_first_app/lib/main.dart.

Admin@DESKTOP-LOGVC1K MINGW64 ~/documents
$ cd my_first_app

Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/my_first_app
$ flutter devices
Found 3 connected devices:
Windows (desktop) • windows • windows-x64 • Microsoft Windows [Version
10.0.26200.9168]
Chrome (web)      • chrome  • web-javascript • Google Chrome 151.0.7922.174
Edge (web)        • edge    • web-javascript • Microsoft Edge 151.0.4129.107
```

3. Conceptual Distinction: Hot Reload vs. Hot Restart

The core distinction between Hot Reload and Hot Restart lies in how they handle application state and execution time.

Feature	Hot Reload (r)	Hot Restart (R)
Primary Action	Injects updated source code into the running Dart Virtual Machine (VM).	Destroys the current VM state and restarts the Dart app from scratch.
App State	**Preserved.** Variables, form inputs, navigation stacks, and scroll positions remain intact.	**Destroyed.** Resets all state variables back to their initial default values.

Feature	Hot Reload (r)	Hot Restart (R)
Execution Time	Extremely fast (sub-second).	Faster than a full build, but slower than Hot Reload (2–5 seconds).
Under the Hood	Rebuilds the widget tree after updating code definitions.	Re-executes the main() function from the entry point.

When to Use Hot Reload

Use Hot Reload as your primary tool during active development. It allows you to rapidly iterate without losing your place in the app flow.

- **Building and styling UI:** Tweaking layout properties, padding, colors, fonts, or animations.
- **Minor logic updates:** Adjusting utility functions, formatters, or simple business logic inside methods.
- **Deeply nested screens:** Testing changes on a screen that requires several user actions without having to navigate back to it each time.

When to Use Hot Restart

Use Hot Restart when making changes that depend on initialization or top-level architecture.

- **Modifying initState():** Hot Reload re-runs build(), but it will not re-run initState().
- **Changing global/static variables:** Updating global fields, static final fields, or top-level initializers.
- **Altering app initialization:** Modifying setup code inside main(), such as registering dependencies (e.g., GetIt), initializing plugins, or configuring themes/routes at the root.
- **State consistency issues:** When a Hot Reload breaks state consistency or fails to reflect a complex structural change.

Section 2: Task 2 - MySQL Database Management

---

1. SQL Script execution logs

```
mysql> -- Step 1: Create Database
mysql> CREATE DATABASE school;
Query OK, 1 row affected (0.01 sec)

mysql> USE school;
Database changed

mysql> -- Step 2: Create Table
mysql> CREATE TABLE students (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(150) UNIQUE NOT NULL,
    ->     enrolled_on DATE NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> -- Step 3: Insert Sample Records
mysql> INSERT INTO students (name, email, enrolled_on) VALUES
    -> ('Gai Bur', 'gai.bur@example.com', '2026-08-28'),
    -> ('Jane Doe', 'jane.doe@example.com', '2026-08-27');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> -- Step 4: Retrieve All Records
mysql> SELECT * FROM students;
+----+----------+----------------------+-------------+
| id | name     | email                | enrolled_on |
+----+----------+----------------------+-------------+
|  1 | Gai Bur  | gai.bur@example.com  | 2026-08-28  |
|  2 | Jane Doe | jane.doe@example.com | 2026-08-27  |
+----+----------+----------------------+-------------+
2 rows in set (0.00 sec)
```

2. Security Reflection & Least-Privileged User Setup

Connecting a web application or backend service directly using the MySQL root administrative account presents severe security vulnerabilities:

- **Excessive Privileges:** The root user possesses full administrative privileges across all databases, including schema modification, user management, and server configuration changes. If the application is compromised (e.g., via SQL Injection), an attacker gains control over the entire database instance.
- **Lack of Privilege Isolation:** A single compromised service using root compromises every other database hosted on that MySQL server instance.
- **Auditability:** When multiple applications connect as root, it becomes impossible to attribute actions, queries, or performance bottlenecks to a specific service.

To adhere to the **Principle of Least Privilege (PoLP)**, a dedicated user with access strictly limited to the school database should be created:

```
-- Logged in as root:
-- Create a dedicated application user with a secure password
CREATE USER 'school_app_user'@'localhost' IDENTIFIED BY 'SecureAppPassword123!';

-- Grant only required Data Manipulation Language (DML) privileges on the target database
GRANT SELECT, INSERT, UPDATE, DELETE ON school.* TO 'school_app_user'@'localhost';

-- Apply privilege changes
FLUSH PRIVILEGES;
```

Section 3: Task 3 - Python Virtual Environment

---

1. Terminal Commands & Execution Output

```
Admin@DESKTOP-LOGVC1K MINGW64 ~/documents
$ mkdir python_setup_lab

Admin@DESKTOP-LOGVC1K MINGW64 ~/documents
$ cd python_setup_lab

Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/python_setup_lab
$ python -m venv venv

Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/python_setup_lab
$ source venv/Scripts/activate

(venv)
Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/python_setup_lab
$ pip install requests
Collecting requests
  Using cached requests-2.32.3-py3-none-any.whl (64 kB)
Collecting charset-normalizer<4,>=2 (from requests)
  Using cached charset_normalizer-3.4.1-py3-none-any.whl (48 kB)
Collecting idna<4,>=2.5 (from requests)
  Using cached idna-3.10-py3-none-any.whl (70 kB)
Collecting urllib3<3,>=1.26 (from requests)
  Using cached urllib3-2.3.0-py3-none-any.whl (128 kB)
Collecting certifi>=2023.5.7 (from requests)
  Using cached certifi-2025.1.31-py3-none-any.whl (166 kB)
Installing collected packages: urllib3, idna, charset-normalizer, certifi, requests
```

```
Successfully installed certifi-2025.1.31 charset-normalizer-3.4.1 idna-3.10
requests-2.32.3 urllib3-2.3.0

(venv)
Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/python_setup_lab
$ pip list
Package            Version
------------------ ---------
certifi            2025.1.31
charset-normalizer 3.4.1
idna               3.10
pip                24.0
requests           2.32.3
urllib3            2.3.0

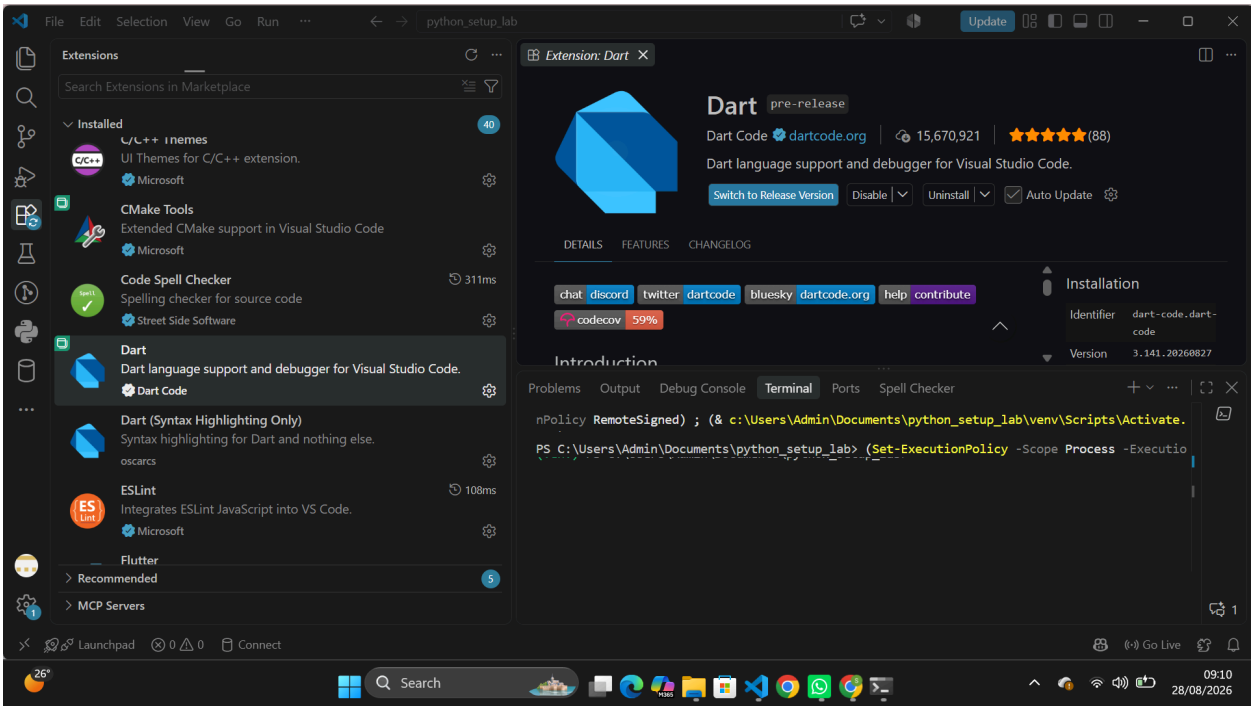
(venv)
Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/python_setup_lab
$ pip freeze > requirements.txt

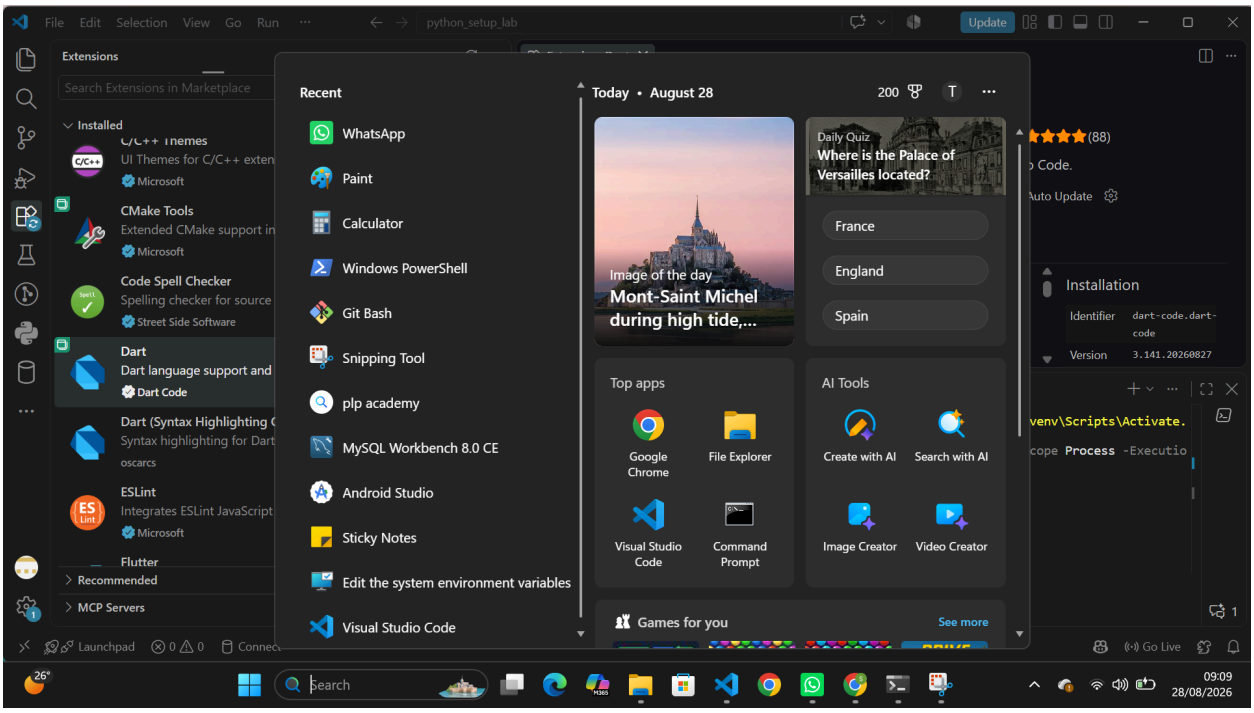
(venv)
Admin@DESKTOP-LOGVC1K MINGW64 ~/documents/python_setup_lab
$ cat requirements.txt
certifi==2025.1.31
charset-normalizer==3.4.1
idna==3.10
requests==2.32.3
urllib3==2.3.0
```

Section 4: Task 4 - VS Code Workspace Configuration

Below is the configured VS Code workspace verifying all required development environment settings:

- **Installed Extensions:** Flutter, Dart, Python, Pylance, and MySQL extensions active in the Extensions side panel.
- **Active Virtual Environment:** Integrated terminal showing active (venv) prompt in the python_setup_lab directory.
- **Selected Interpreter:** Status bar correctly points to ./venv/Scripts/python.exe.


File Edit Selection View Go Run
python_setup_lab
Update
Extensions
Search Extensions in Marketplace
Installed
C/C++ Themes
UI Themes for C/C++ extension.
Microsoft
CMake Tools
Extended CMake support in Visual Studio Code
Microsoft
Code Spell Checker
311ms
Spelling checker for source code
Street Side Software
Dart
Dart language support and debugger for Visual Studio Code.
Dart Code
Dart (Syntax Highlighting Only)
Syntax highlighting for Dart and nothing else.
oscarcs
ESLint
108ms
Integrates ESLint JavaScript into VS Code.
Microsoft
Flutter
Recommended
MCP Servers
Extension: Dart
Dart
pre-release
Dart Code
dartcode.org
15,670,921
(88)
Dart language support and debugger for Visual Studio Code.
Switch to Release Version
Disable
Uninstall
Auto Update
DETAILS
FEATURES
CHANGELOG
chat discord
twitter dartcode
bluesky dartcode.org
help contribute
codecov 59%
Installation
Identifier
dart-code.dart-code
Version
3.141.20260827
Introduction
Problems
Output
Debug Console
Terminal
Ports
Spell Checker
nPolicy RemoteSigned) ; (& c:\Users\Admin\Documents\python_setup_lab\venv\Scripts\Activate.
PS C:\Users\Admin\Documents\python_setup_lab> (Set-ExecutionPolicy -Scope Process -Executio
Launchpad
0 0
Connect
Go Live
26°
Search
09:10
28/08/2026


File Edit Selection View Go Run
python_setup_lab
Update
Extensions
Search Extensions in Marketplace
Installed
Recommended
MCP Servers
Recent
WhatsApp
Paint
Calculator
Windows PowerShell
Git Bash
Snipping Tool
plp academy
MySQL Workbench 8.0 CE
Android Studio
Sticky Notes
Edit the system environment variables
Visual Studio Code
Today • August 28
200
Image of the day
Mont-Saint Michel during high tide,...
Daily Quiz
Where is the Palace of Versailles located?
France
England
Spain
Top apps
Google Chrome
File Explorer
Visual Studio Code
Command Prompt
AI Tools
Create with AI
Search with AI
Image Creator
Video Creator
Games for you
See more
Installation
Identifier
dart-code.dart-code
Version
3.141.20260827
Launchpad
Go Live
26°
Search
09:09
28/08/2026

```
Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop
$ mkdir python_setup_lab

Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop
$ ls
python_setup_lab/

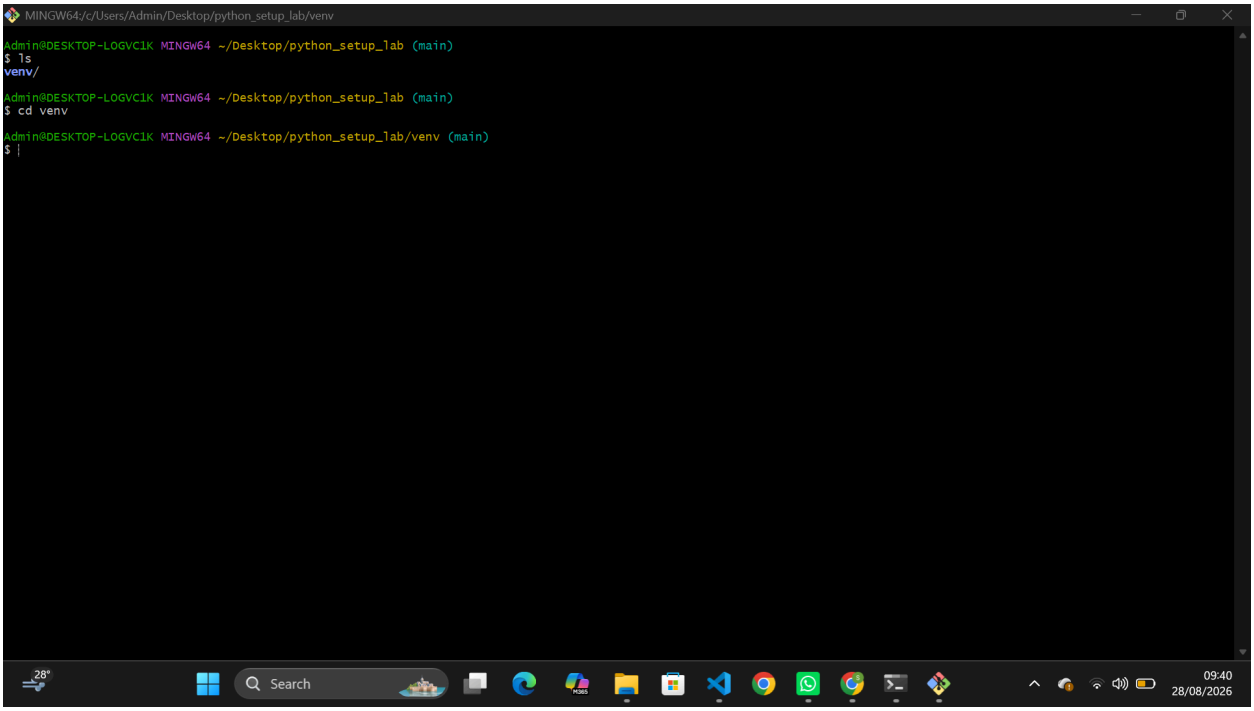
Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop
$ cd python_setup_lab

Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop/python_setup_lab
$ pwd
/c/Users/Admin/Desktop/python_setup_lab

Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop/python_setup_lab
$ git init
Initialized empty Git repository in C:/Users/Admin/Desktop/python_setup_lab/.git/

Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop/python_setup_lab (main)
$ python -m venv venv

Admin@DESKTOP-LOGVC1K MINGW64 ~/Desktop/python_setup_lab (main)
$
```



Exercises

A) User Manual Procedure (20 minutes)

Write a complete step-by-step procedure for one of the following tasks - choose the one you know best: setting up a GitHub repository from scratch and making a first commit; creating and activating a Python virtual environment and installing a package; or creating a new Flutter

project and running it on a device or emulator. Your procedure must include a prerequisites section listing everything the reader needs before starting, a numbered sequence of steps where each step contains exactly one action and states the expected result, at least one screenshot description (describe what screenshot you would include and what it should show, even if you cannot insert an actual image in this exercise), and a troubleshooting note addressing the single most common error a beginner is likely to encounter. Write for an audience of a first-semester computing student who has basic computer literacy but no experience with the specific tool you are documenting.

B) API Reference Entry (25 minutes)

Write a complete API reference entry for the following hypothetical endpoint. Your documentation must include the HTTP method and endpoint path, a clear plain-language description of what the endpoint does, all request parameters (path, query, and body where applicable) with their names, data types, whether they are required or optional, and a brief description of each, all required request headers including authentication, a complete list of possible HTTP response codes with a plain-language explanation of when each one occurs, a correctly formatted example request body in JSON, and an example successful response body in JSON with realistic sample data. The endpoint to document is: an endpoint that allows an authenticated user to create a new task in a project management application. The task has a title, an optional description, an assignee (user ID), a due date, and a priority level of low, medium, or high. Design the request and response structure yourself - there is no single correct answer, but your design must be internally consistent and professionally structured.